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Interleaving Rhythmic and Discrete Control for the Manipulation of Underactuated Objects

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Abstract— **Humans frequently manipulate and move dynamically complex objects, e.g., when swirling and drinking a glass of wine. In robot control, dynamic motion primitives have been adopted as control components to generate complex behaviors. How the human motor system employs discrete and rhythmic movement primitives when manipulating underactuated dynamical systems remains unknown. Inspired by the task of swirling a glass of wine, we designed a virtual task in which participants moved a cup containing a sliding ball on a horizontal surface. Participants first moved the cup to steadily swirl the ball, then translated the cup to a target location while maintaining ball swirl. Performance was quantified by the variability of the ball's radial distance from the rim during the translation phase. Adopting a task-dynamic approach, we found that participants consistently initiated the translation when the ball angle was within a specific angular window, although the translation durations varied across individuals. Optimal control simulations revealed that selective combinations of ball initial angle and translation duration naturally achieved stable ball swirl during translation. The experimentally observed ball states at translation onset and the corresponding translation times closely matched these optimal combinations. These results gave support that humans coupled rhythmic and discrete movement components to simplify control and stabilize dynamics of underactuated objects.**

I. INTRODUCTION

Humans are capable of generating highly skilled and versatile movements across a wide range of behaviors with dynamically complex objects, from tying shoelaces to carrying a cup of coffee. In such complex interactions, the movement goals and the foci of control can be multi-layered. When bringing a glass of wine to one's mouth, the primary focus is on the controlled object, the glass, that needs to be translated towards the mouth. In contrast, when swirling a glass of wine, the primary focus is on the wine, the internal degree of freedom, to bring it into a rhythmic circular motion. In many daily activities, rhythmic and discrete motions are intertwined. This human capability to flexibly create and intertwine such complex motions still remains a benchmark for state-of-the-art robots.

One promising proposition to understand the control of such complex actions is that movements emerge from the combination of simpler movements, or primitives [1, 2, 3, 4]. Humans then employ and merge such movement primitives to produce goal-directed behavior. Studies in motor neuroscience have provided robust evidence for the existence of rhythmic pattern generators, primarily for locomotory behaviors in animals [5, 6], but also in upper-limb behaviors in humans [7, 8]. Behavioral and electromyographic studies also revealed evidence for discrete pattern generators as fixed-point attractors in the form of synergies, again in both animals and humans [9, 10].

Due to their elegant and simple formulation, dynamic motion primitives (DMPs) have already been used extensively in robotics to enable robots to perform complex tasks [11, 12]. However, it still remains unclear whether humans combine discrete and rhythmic movement primitives, especially in more interactive tasks like manipulating objects with underactuated dynamics. Understanding human control could further inform the development of robotic control strategies, specifically for manipulating flexible and underactuated objects, which continues to pose a significant challenge.

Prior studies on rhythmic and discrete interactions have primarily examined single-joint or two-joint tasks [3, 13, 14]. These studies demonstrated that discrete and rhythmic primitives are not combined by simple superposition or sequential execution. Instead, the onset of the discrete movement is confined to a specific phase window of the rhythmic movement, even when the discrete movement is to be produced as quickly as possible. Reaching movements are initiated at the peak momentum of the muscle-joint system, even when the oscillation is only caused by subtle physiological tremor rhythms [15]. Taken together, these findings indicate that the two movement primitives mutually influence one another. This present work investigated such inter-dependencies when humans control an object with underactuated dynamics.

Manipulating objects with underactuated dynamics requires more subtle movements than when moving rigid objects as the human has to contend with complex interaction forces. For example, prior work has shown that the characteristic minimum-jerk trajectories observed in human reaching with rigid objects are no longer seen in movements involving underactuated mass–spring systems [16, 17, 18]. When moving a cart-pendulum system, combining two appropriately scaled and timed discrete primitives can simplify control, and mitigate pendulum oscillation at the final location [19, 20].

A significant challenge when manipulating complex dynamical objects is ensuring dynamical stability. For instance, during rhythmic manipulation of a cup with a ball sliding inside, the ball may exhibit large oscillations and potentially escape from the cup. Our previous work has shown that hu-

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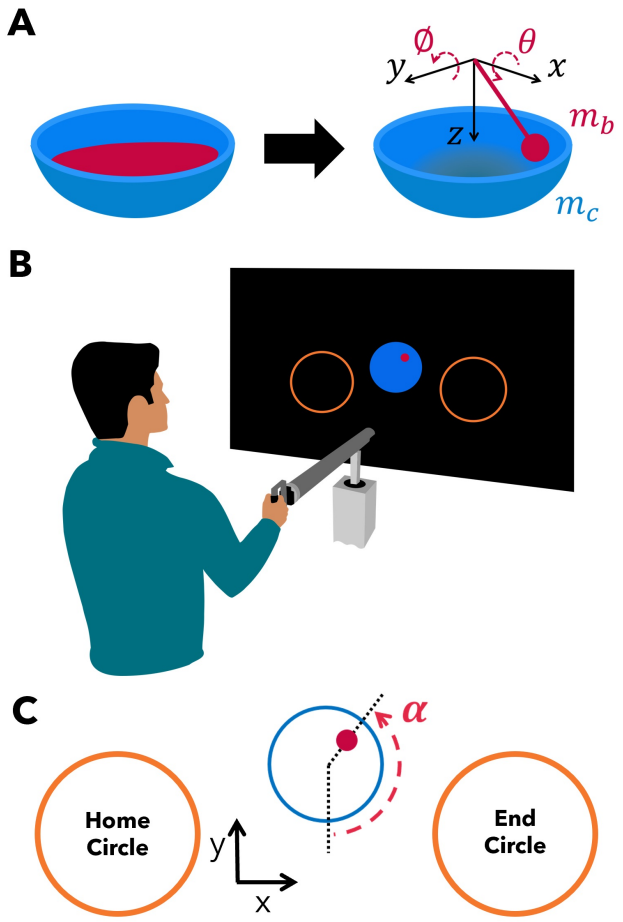


Fig. 1: **Experimental Task and Protocol. A.** Wine sloshing inside a cup was simplified to a ball inside the cup, and modeled as a spherical pendulum on a cart. **B.** Participants first swirled the ball maintaining the cup within a home circle, translated the cup into a end circle while maintaining ball swirl, and continued swirling the ball while keeping the cup inside the end circle. **C.** Participants controlled the cup-ball system displayed on a vertical screen by moving the handle of a robotic manipulandum in the horizontal plane. The cup position was directly mapped to the location of the handle.

mans select interaction frequencies to ensure stable dynamics of such a cup-ball system manipulated in one dimension [21, 22]. Further, humans prepare initial conditions of the cup-ball system before executing rhythmic back-and-forth movements of the cup to ensure stable dynamics of the ball [23], [24]. These studies primarily focused on either rhythmic or discrete manipulation. However, manipulation of complex dynamical objects often also involves both discrete and rhythmic components, as in the example of moving a glass of wine while swirling the wine. To maintain the dynamical stability of the object, rhythmic and discrete movement variables likely need to be subtly coupled.

The present work adopted an object- or task-dynamic approach to investigate how discrete and rhythmic movements are coupled when humans manipulate objects with underactuated dynamics. The task, inspired by swirling a glass of wine, was simplified to controlling a ball sliding inside a cup. Participants had to establish a steady swirling motion of the ball before translating the cup to a new location while maintaining the ball's swirl. Optimal combinations of the initial ball state at translation onset with translation durations could preserve ball stability during translation. Model simulations demonstrated that humans may simplify control with dynamic primitives to ensure stability when manipulating objects with underactuated dynamics.

II. METHODS

A. Participants

Seven college students participated in this experiment, including 3 females and 4 males. All participants were right-handed, had no reported biomechanical issues in their upper limbs, and were unaware of the purpose of the study. They signed an informed consent form approved by the Northeastern University Institutional Review Board prior to participation.

B. Experimental Apparatus

We simplified the task of swirling wine in a glass to moving a cup while controlling a ball sliding inside close to the circular rim (Fig.1A,B). The cup-ball system was mechanically equivalent to a spherical pendulum suspended from a cart, where the cart and the pendulum bob represented the cup and ball, respectively. The cup position was defined in two dimensions x and y, and the ball movements were defined using Euler rotations θ and ϕ around the x and y axis, respectively. The following dynamical equations describe the cup-ball system:

$$M\ddot{q} + C(\dot{q},\ q) + G = F \tag{1}$$

where, q is the state vector $[x,\ y,\ \theta,\ \phi]$. $F = [F_x,\ F_y,\ 0,\ 0]$ with F_x and F_y denoting the forces applied to the cup in the x and y directions, respectively. M, C and G are matrices defined as follows:

$$M = \begin{bmatrix} m_c + m_b & 0 & m_b l S(\phi) S(\theta) & -m_b l C(\phi) C(\theta) \\ 0 & m_c + m_b & m_b l C(\theta) & 0 \\ m_b l S(\phi) S(\theta) & m_b l C(\theta) & m_b l^2 & 0 \\ -m_b l C(\phi) C(\theta) & 0 & 0 & m_b l^2 S^2(\theta) \end{bmatrix} \tag{2}$$

$$C = \begin{bmatrix} m_b l S(\phi) C(\theta)(\dot{\theta}^2 + \dot{\phi}^2) + 2 m_b l \dot{\phi}\dot{\theta} C(\phi) S(\theta) \\ -m_b l S(\theta)\dot{\theta}^2 \\ b_b\dot{\theta} + m_b l^2 S(\theta) C(\theta)\dot{\phi}^2 \\ b_b\dot{\phi} + 2 m_b l^2 S(\theta) C(\theta)\dot{\phi}\dot{\theta} \end{bmatrix} \tag{3}$$

$$G = \begin{bmatrix} 0 \\ 0 \\ m_b l g S(\theta) C(\phi) \\ m_b l g C(\theta) S(\phi) \end{bmatrix} \tag{4}$$

where $S(\cdot)$ and $C(\cdot)$ represent sine and cosine trigonometric functions, respectively. The mass of the cup was m_c = 2.58 kg, the mass of the ball was m_b = 0.681 kg, and the gravitational acceleration was g = 9.81 m/s^2. The length of the pendulum was l = 0.052 m and the viscous damping at the pendulum pivot was b_b = 0.004 Ns/m.

C. Experimental Task and Procedures

The task was implemented in a virtual environment and displayed on a large projection screen facing the participant. The cup-ball system was displayed in a top-down view with the cup shown as a blue solid disc (radius = 7.1 cm) and the ball as a red solid disc (radius = 1.9 cm) (Fig. 1B,C). Participants sat on a chair in front of the screen and gripped the handle of an admittance-controlled robotic manipulandum (HapticMaster, Motekforce, Netherlands, Fig. 1B). The motion of the robotic handle was constrained to the horizontal plane. By moving the robotic handle, participants controlled the (x, y) position of the cart/cup directly. Two open orange circles (radius = 9 cm) were displayed on the screen, representing the home and end circles.

The position of the chair was adjusted such that the participant could comfortably grasp the handle and move the cup between the two circles. Forces from the cup-ball model were fed back to the participant's hand via the robotic handle. The x and y positions of the handle (i.e, cup position), the positions and Euler angles of the computed and virtually displayed ball motions were recorded at 600 Hz using a custom-written C++ program.

At the beginning of each trial, the cup and ball were stationary inside the home circle. Participants then started to move the cup and swirl the ball into a steady rotation, either in the clockwise (CW) or the counterclockwise (CCW) direction, while maintaining the cup within the home circle (Fig. 2A). Once the participant settled into steady ball swirling, a go cue was displayed on the screen. The participant could then initiate the translation movement whenever they were ready. During the translation to the end circle, they were required to maintain the ball rotations. Upon reaching the end circle, they continued swirling the ball, while keeping the cup within the circle. The trial ended 3 s after arriving at the end circle. The trial was considered unsuccessful, if the ball went beyond the rim of the cup at any point during the trial. If the trial was not successful, it was repeated.

In summary, each trial had three stages i) swirl prior - rotating the ball at the home circle, ii) translation - moving the cup to the end circle while maintaining the ball swirl, and iii) swirl post - continuing to swirl the ball at the end circle. Successive trials alternated between starting at the home circle and starting from the end circle. All participants completed a total of 400 successful trials over three days (100 trials on Day 1, and 150 trials on Days 2 and 3). On an average participants had around 60 unsuccessful trials for 400 successful trials.

D. Data Analysis and Dependent Measures

Position and force data of all trials were mapped to one movement direction, with both the home circle and the end circle located on the positive x-axis. To pool all trials, clockwise rotations were transformed to counterclockwise rotations. All data were low-pass filtered using a 4th-order Butterworth filter with a 10 Hz cut-off frequency.

1) Translation Duration: Starting from the moment of the peak cup velocity in the x-direction, we moved backward or forward in time to identify the times at which the velocity crossed a threshold 5% of the peak velocity. These two time points were defined as translation start and end, respectively. The difference between the end and start times defined the translation duration.

2) Cup and Ball Oscillation Period: The peak-to-peak times of the cup positions were calculated together with the Euler angles of the ball. The average peak-to-peak times in the x and y positions were denoted as the cup oscillation period. The average peak-to-peak times of the θ and ϕ angles were denoted as the ball oscillation period.

3) Ball Swirl Stability: The ball swirl radius was calculated as the distance of the ball to the center of the cup. The standard deviations of the ball swirl radius captured a participant's ability to maintain stable swirl of the ball. We were specifically interested in the variability of the ball swirl radius during the more challenging translation stage, that we refer to as ball swirl stability.

4) Ball Azimuthal Angle: Ball azimuthal angle α was the angular position of the ball around the z-axis (Fig.1A) passing through the center of the cup.

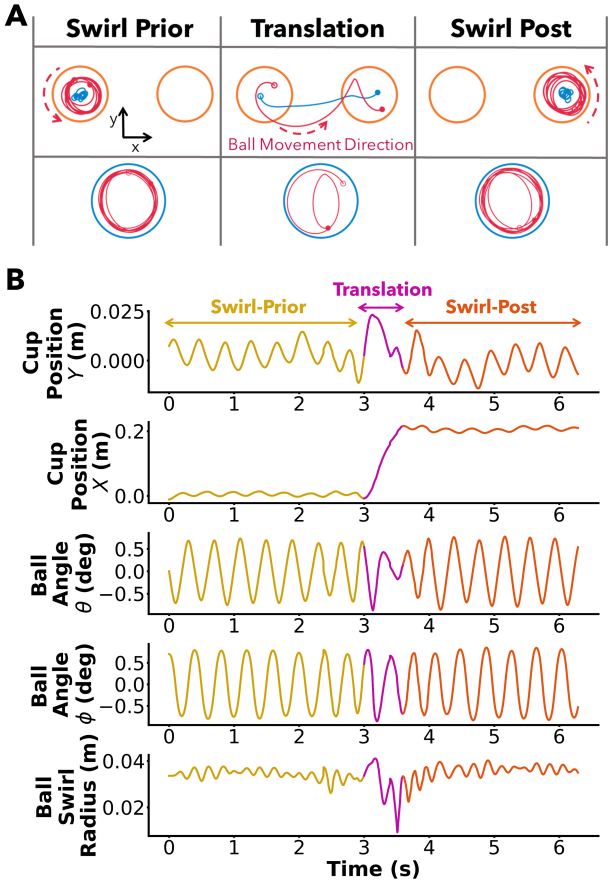


Fig. 2: **Cup and Ball Kinematic States Example Trial**. **A.** Two-dimensional trajectories of the cup and ball are shown for an example trial in the top row. The bottom row shows a ball trajectory relative to the cup. **B.** Corresponding cup positions, ball Euler angles, and ball swirl radius as a function of trial time.

E. Task-Dynamic Analysis and Optimal Control Simulations

To identify solutions to the dynamic task, we started with the physical model of the object and then identified "execution variables", defined as variables that fully determined the object's motions in the task-specified fashion. For the present task, ball angle and translation duration were two essential variables that fully determined the resulting stability of the object. Stability of the ball motion was chosen as the "result variable", the variable that resulted from manipulating the system through the execution variables, consistent with the instruction to the subject. Stability was operationalized by the variability of the ball motions. Assuming rhythmic ball motions (rhythmic primitive) and a discrete cup movement (discrete primitive), an optimal controller was used to create a mapping between execution and result variables, the task's solution space.

The cost function for the optimal controller was designed to emulate the task instructions in the three task stages. The desired ball oscillation period and ball radius for the swirl-prior and swirl-post stages were taken from the average experimental data to be 0.4 s and 3.2 cm, respectively. The ball's Euler angle trajectories were defined accordingly. The desired cup position was stationary at the home and end locations during the prior and post-swirl stages, respectively. The gain for tracking the ball trajectory was set higher than the gain for keeping the cup stationary at the home or end circles (3:1 ratio). The state cost matrix was designed accordingly to achieve the required cup and ball behavior.

It is important to point out that there was no cost included to maintain ball swirl during the translation. Thus, the standard deviation of the ball swirl radius during the translation was the result variable that arose from the cost function. Costs were also imposed on the magnitude of the applied force to limit effort. The time step for the simulations was 0.01 s. The total simulation time was 5 s. The cost function could be summarized as follows:

$$J = \sum_{0}^{t_s} q^T Q_{rhythm} q + \sum_{t_s}^{t_e} q^T Q_{discrete} q + \sum_{t_e}^{500} q^T Q_{rhythm} q + \sum_{0}^{500} F^T R F \quad (5)$$

where t_s is the start time of translation, and t_e is end time of translation; Q_{rhythm} is the state cost matrix for the swirl stages; $Q_{discrete}$ is the state cost matrix for the translation stage; R is the control cost matrix for the entire simulation trial; q and F are state and force input vectors defined in Eq.1.

1) Simulations: Differential dynamic programming was used to iteratively find control increments that minimized the cost function [25, 26]. The start of the translation t_s and the translation duration $(t_e - t_s)$ of the cup were varied between simulation runs based on the ball azimuthal angle. A ball oscillation cycle between 2 s and 2.4 s was chosen, respectively corresponding to ball azimuthal angles 0 deg and 360 deg. The start of the translation t_s was defined at any given ball azimuthal angle within the chosen time window. Between simulation runs, the ball azimuthal angle was discretized between the bounds $(0,\ 360)$ deg in steps of 10 deg. The translation duration was discretized between $(0.2,\ 0.9)$ s in steps of 0.03 s. We ran simulations for all combinations of ball azimuthal angle and translation durations.

To quantify the overlap between the participants' data distributions with the regions of stability generated by the model we used the Jensen-Shannon (JS) divergence metric. A high value of JS-divergence indicates non-overlapping distributions, while low JS-divergence value indicates a good match. To afford the comparison of the participants' data distributions with the model-generated stability regions, the discrete participant data were converted to a polar probability distribution with the same level of discretization as the heatmap of the model-generated data. The model is supported if the overlap is high or the JS-divergence values were low. In addition, we compared the participants' data distributions to heatmaps generated by randomly permuted stability values of the model (compiled over 100 permutations). If the model-generated explained participant data better, the JS-divergence should be lower than the one obtained from the comparison with the permuted model heatmap.

III. RESULTS

A. Cup and Ball Kinematics

Cup and ball positions from an example trial are shown for the three different stages in Fig. 2A. The top row shows cup and ball motions in x-y coordinates, while the bottom row shows ball trajectories in cup coordinates. Fig.2B illustrates the time series of the cup x and y, ball Euler angles, and ball swirl radius. The cup position in y shows the rhythmic

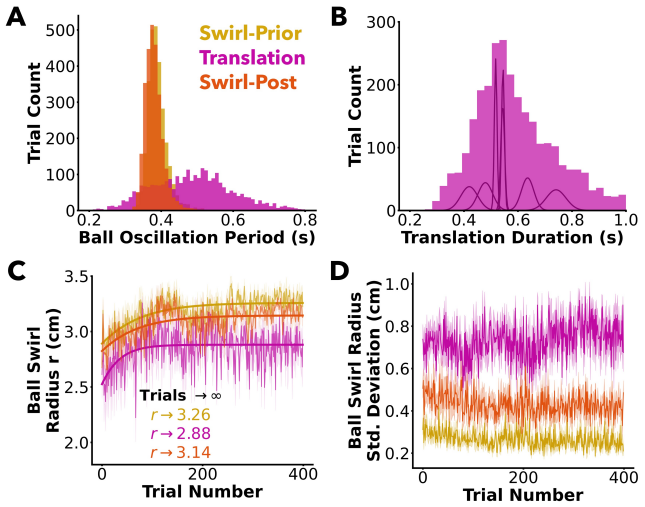


Fig. 3: **Ball Oscillation Period and Swirl Radius**. **A.** Histogram of ball oscillation period from all trials and participants plotted separately for the three trial stages. Ball oscillation periods were concentrated around 3.9 s for the swirl-only stages. **B.** Histogram of translation durations from all participant. Dark lines represent individual distributions. **C.** Mean ball swirl radius across participants plotted against trials. Exponential fits indicate ball swirl radius converged to a value of 3.2 cm in the swirl-only stages. **D.** Mean of standard deviation of ball swirl radius across participants plotted across trials for the three trial stages. All error ribbons show ± 1 standard error between participants.

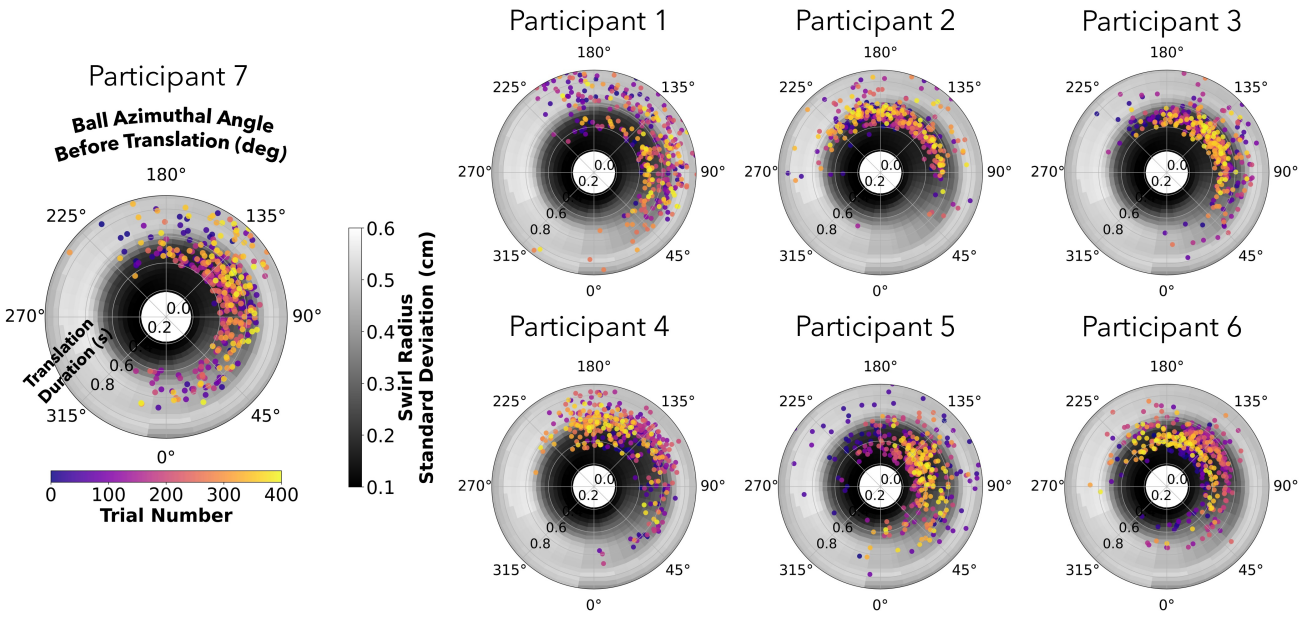


Fig.4: **Task Solution Space**. Heatmap of standard deviation of swirl radius (stability of swirl radius) plotted for combinations of ball azimuthal angle at the start of translation and translation duration for one participant on the left. Darker regions of the heatmap imply lower standard deviation, i.e., higher stability. Experimental data for the participant from 400 trials is plotted as circles on top of the heatmap. The circles are colored according to the trial number. Similar plots for the remaining participants are shown on the right.

component during the cup translation. The ball Euler angles indicate that the ball continued to swirl during the translation of the cup and also slightly affected the cup position.

Ball oscillation periods from all participants' trials are summarized in the histograms in Fig.3A. The periods in the swirl-only stages were concentrated around 0.39 s with a standard deviation of 0.026 s. The translation durations were distributed between 0.3 s and 1.0 s (Fig.3B). Selected individual distributions of translation durations, drawn as dark lines, show that individuals differed quite substantially in their preferences. Fig.3C displays ball swirl radius from all participants plotted across trials. The ball radius was lowest during the translation stage compared to swirl-prior and swirl-post, where it converged to a value around 2.9 cm. The standard deviation of the ball swirl radius was highest during the translation, lowest in the swirl-prior stages, and marginally improved in the swirl-only stages (Fig.3D).

B. Solution Space and Experimental Data

Fig.4 shows the solution spaces generated by the optimal control simulations: the polar heatmaps plot the standard deviation of the ball swirl (result variable, stability) against ball azimuthal angle and translation duration (execution variables). Darker regions indicate lower standard deviation of the ball swirl radius. It is important to point out again that the optimal controller did not explicitly strive to maintain ball swirl during the translation. However, appropriate selections of ball azimuthal angle at the start of translation and the translation duration naturally ensured lower standard deviations.

The polar maps show higher ball stability for shorter cup translations with nonlinear changes across durations. A faint spiral pattern could be discerned that indicated the interrelation between the azimuth angle and the translation duration. The area of stable ball swirl stretched over a larger range of durations for ball angles between 90 and 180 deg.

The experimentally observed ball azimuthal angles and translation durations from all 400 trials from one participant are plotted over the model heatmap, separately for all seven participants. Most participants chose the ball azimuthal angle and translation durations that ensured lower standard deviations of ball swirl radius during translation. To examine learning effects, the data points are colored according to the trial number. Qualitatively, they did not appear to converge to the more stable regions of the heatmap, i.e., they did not show changes with practice.

Fig.5 quantifies and summarizes the Jensen-Shannon divergence between the participant data distributions and the underlying model heatmap. The red points represent the JS-divergence for the seven participants with the model-generated heatmap. For reference, the boxes represent the JS-divergence between the data probability distributions and the randomly permuted model heatmaps (produced by 100 permutations, plotted each participant. The JS-divergence values between the model heatmap and empirical data were overall lower compared to the permuted heatmap, indicating that the model explained participant data better than the permuted heatmap. Exceptions are Participants 1 and 4.

IV. DISCUSSION

This study investigated a complex human manipulation task that interleaved discrete and rhythmic elements. Participants controlled a ball in a cup while also moving the cup from one point to another. Per instruction, participants rotated the ball inside the cup before, during and after a discrete translation of the cup. Results showed that participants were

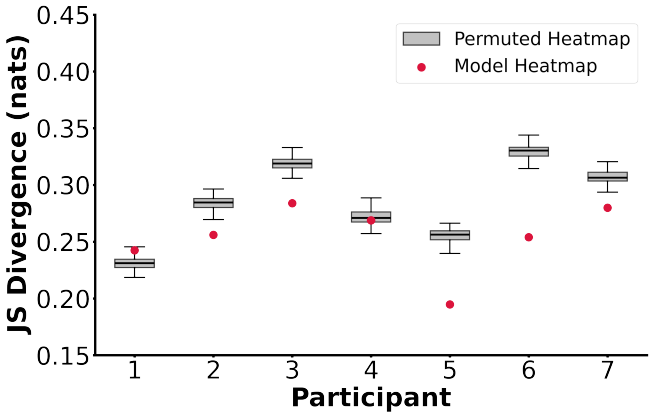


Fig.5: **Quantitative Comparison between Empirical Data and Model Heatmap**. Jensen-Shannon divergence between participant data distributions and the model-generated heatmap are shown for the seven participants as red points. JS-divergence values between the participant data distribution and randomly permuted model distributions for 100 runs are shown as box plots.

able to maintain stable ball swirl with consistent oscillation periods and swirl radii, although the ball rotations were more variable during the translation.

How can one shed light on the control of this complex object with multiple intertwined elements? Our previous work had developed a task-dynamic approach where the analysis begins with the physical model of the object [21, 27, 28]. The critical step is to identify the execution variables that are essential in creating the desired object's behavior that is defined by one or more result variables. Once execution and result variables are defined, a solution space can be created by simulating the object's model activated by the execution variables. This solution space shows areas of successful results derived from this object-centered task analysis. Our underlying hypothesis is that control of the task is generated via a few execution variables that engage dynamic primitives, specifically rhythmic and discrete building blocks [4, 29].

This paper aimed to exemplify how the complex wine-swirling task can be understood by this task-centered approach. We showed how rhythmic and discrete movements that, when integrated appropriately, can ensure stable performance, specifically for our task the relatively stable ball rotations. The result variable stability was operationalized by standard deviation of the ball radius, in accordance with the instruction. Ball azimuthal angle and translation duration were identified as the critical execution variables that had to be coupled to achieve the best result. The generated solution space revealed the specific combinations of the ball's azimuthal angle and translation duration that ensured the ball's dynamic stability. Human data showed congruence with the hypothesized coupling of the two variables.

All participants established relatively steady ball oscillations during the swirl stages by subtly oscillating the cup. Prior studies have shown that the input frequency to a nonlinear system influenced the stability of underactuated dynamics [21, 30]. One study of our group on cup-ball rhythmic manipulation in 1D showed that participants converged to cup frequencies that ensured predictability and stability of the ball dynamics with respect to the cup movements [21]. Hence, in our 2D cup movements participants potentially converged to specific cup and ball oscillation periods to enhance stability of ball swirl dynamics. Additionally, the ball oscillation periods could have been selected to control the swirl radius to ensure sufficient safety margins to prevent the ball from escaping beyond the rim of the cup.

A significant challenge was to maintain ball swirl during translation of the cup and at the end circle. Although the ball radius was less stable during the translation, the ball swirl was sufficiently sustained. The x position of the cup, i.e., in the direction of the location of the end circle, resembled a typical point-to-point displacement, consistent with the assumption of a discrete primitive. In contrast, the y position of the cup exhibited rhythmic components to maintain ball swirl, consistent with an oscillatory primitive. Further analysis is necessary to test the integration or dissociation of control strategies in orthogonal movement dimensions.

Another strategy that can be used to ensure stable ball swirl during translation is to covary rhythmic and discrete movement features. Our optimal control modeling results indicated that specific combinations of the ball state at the beginning of translation and the translation duration naturally ensured stable ball swirl during translation. Indeed, the experimental data showed that participants adopted such a control strategy. It would be interesting to test how enforcing costs on maintaining ball swirl during translation affects stability of ball swirl and effort expenditure. Inspired by central pattern generators, past studies used nonlinear dynamical models to generate discrete and rhythmic movement primitives and combined them to produce complex single-joint movements [3, 11, 31, 32]. Synchronization between the oscillator and discrete primitives predicted entrainment of the initiation of the discrete movement to the phase of the rhythmic movement [13]. Our findings replicated these results in the more complex scenario of manipulating an underactuated system. Our past work on complex object manipulation also highlighted the importance of preparing object states to ensure stable dynamics [22, 23, 33]. Initiation of translation at a specific ball angle is consistent with this emphasis on how preparation helps to establish stability.

The object-centered analysis of this task highlighted how understanding the object's dynamics constrains the solutions and reveals possible control strategies for this complex task. Based on the characterization of the task's affordances, simple building blocks, discrete and rhythmic primitives, may be selected optimally to ensure stable dynamics. Such a solution would circumvent or attenuate costly computation required for feedback-based control of the ball. This is consistent with and also extends previous work on dynamic movement primitives (DMPs) as a viable avenue to elegantly design robotic control policies [1, 12]. Some robotic studies have sequenced movement primitives using thresholds [34], or used overlapping basis functions at transition points [35]. Our results point to the possibility that the coupling of forcing terms of the movement primitives can be co-optimized to generate simplified control policies. One challenge remains

that the solution landscape is likely to change with changes in the imposed translation timing or the amplitude of the ball swirl. Future work needs to address how the solution space changes with different imposed rhythmic and discrete movement features.

V. CONCLUSIONS

Our results reveal motor control strategies that humans employ when combining discrete and rhythmic movement primitives to manipulate objects with nonlinear underactuated dynamics. Physical interaction with objects exhibiting such complex dynamics remains a significant challenge for modern robotic systems. Using human experimental data and optimal control simulations, we demonstrate how discrete and rhythmic movement variables can covary to simplify control and stabilize underactuated dynamics. These findings suggest that human movement strategies provide valuable insights for the design of robotic controllers. Future research may focus on developing learning frameworks that naturally yield such simplified control policies.

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